

EXPERIMENT - 4

Data Modelling and Form Design for a Business Application in Zoho Creator

Vehicle Service & Repair Request Management

Application Name	Vehicle Service & Repair Request Management
Form Name	Service_Request_Form
Report Name	Service_Request_Report
Student Roll No.	241401020
Experiment	4
Platform	Zoho Creator

Lab Record

A low-code business application designed to capture, validate, schedule, and track vehicle service and repair requests using all thirteen Basic Fields available in the Zoho Creator Form Builder.

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1. Objective

To design and build a functional Vehicle Service & Repair Request Management application in Zoho Creator that demonstrates the correct usage of all thirteen Basic Fields in the Form Builder while addressing the practical needs of an automobile service center.

2. Problem Statement

The service center traditionally receives requests through phone calls. This can cause double-booked service bays, incomplete information about the reported problem, delays in preparing parts or estimates, and long waiting times for walk-in customers. A structured online request form provides advance information and enables the front office and technicians to prepare before the vehicle arrives.

3. Scope of the Use Case

Application	Vehicle Service & Repair Request Management
Form	Service_Request_Form
Primary Users	Vehicle owners, Service Center Front Office, Technicians
Report View	Service_Request_Report
Report Objective	Filter incoming requests by service type/date and support technician & bay assignment

4. Business Data Model

The form acts as the primary data-entry interface. Each field maps to one business attribute of a service request. The resulting record becomes a row in Service_Request_Report, allowing the service center to review and schedule work.

Core entity: Service Request → Owner details + Vehicle details + Contact details + Issue details + Schedule + Authorization.

5. Basic Field Mapping - All 13 Fields

The following design uses every Basic Field listed in the exercise and assigns each one a clear business purpose.

#	Basic Field	Form Field Name	Purpose / Example
1	Name	Owner_Name	Owner's first and last name; e.g., Ramesh Kannan
2	Email	Email_Address	Validated email for estimate and status updates
3	Address	Pickup_Address	Pickup/drop address when service is requested
4	Phone	Contact_Number	Mobile number for service-center communication
5	Single Line	Vehicle_Registration_Number	Short vehicle registration / number plate
6	Multi Line	Issue_Description	Detailed description of the vehicle problem
7	Number	Odometer_Reading	Current odometer reading in kilometres
8	Date	Preferred_Service_Date	Preferred service date
9	Time	Preferred_Dropoff_Time	Preferred drop-off time slot
10	Drop Down	Vehicle_Type	Single vehicle category such as Hatchback
11	Radio	Service_Type	Routine / Breakdown Repair / Accident Repair
12	Multi Select	Additional_Services_Requested	Car Wash, Interior Cleaning, Tyre Rotation
13	Checkbox	Authorize_Estimate_And_Repair	Customer authorization to prepare estimate and proceed

6. Section-wise Form Design

Section A - Owner Details	Owner_Name; Vehicle_Registration_Number; Vehicle_Type
Section B - Contact Details	Email_Address; Contact_Number; Pickup_Address
Section C - Vehicle & Issue Details	Odometer_Reading; Service_Type; Additional_Services_Requested; Issue_Description
Section D - Scheduling & Authorization	Preferred_Service_Date; Preferred_Dropoff_Time; Authorize_Estimate_And_Repair

7. Validation Rules

Field	Validation / Configuration
Owner_Name	Mandatory; captures first and last name.
Contact_Number	Mandatory; phone field with Indian country-code format.
Vehicle_Registration_Number	Mandatory; short text entry.
Vehicle_Type	Mandatory; one value from the drop-down list.
Service_Type	Mandatory; exactly one radio option.
Authorize_Estimate_And_Repair	Mandatory; form submission is blocked unless checked.
Email_Address	Built-in email format validation.
Odometer_Reading	Number field; positive values only.
Preferred_Service_Date	Date field; past dates are disallowed.

8. Workflow Summary

1. Owner opens the public Service_Request_Form link shared by the service center.
2. Owner enters identity, contact, vehicle, issue, schedule and authorization details.
3. Zoho Creator validates mandatory fields and field formats before submission.
4. A new service-request record is created in Service_Request_Report.
5. An acknowledgement email can be triggered using Email_Address.
6. Front-office staff review the report and filter requests by Service_Type and Preferred_Service_Date.
7. Technicians and service bays can be planned before the vehicle arrives.

9. Sample Business Record

Attribute	Sample Value
Owner_Name	Ramesh Kannan
Email_Address	ramesh.kannan@example.com
Contact_Number	+91 93450 66778
Vehicle_Registration_Number	TN 34 AB 5678
Vehicle_Type	Hatchback
Odometer_Reading	45230 km
Service_Type	Breakdown Repair
Issue_Description	Unusual grinding noise from the front brakes and reduced mileage over the last week
Preferred_Service_Date	30-Jul-2026

Attribute	Sample Value
Preferred_Dropoff_Time	9:00 AM
Additional_Services_Requested	Car Wash, Interior Cleaning, Tyre Rotation
Authorization	Checked (Agreed)

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10. Implementation Evidence

The screenshot displays the Zoho Creator Form Builder interface for a form titled "Vehicle Service & Repair Request Management" with the subtitle "Service_Request_Form". The interface is divided into three main sections:

- Basic Fields:** A sidebar on the left containing various field types categorized by icons: Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, Drop Down, and a "More" option.
- Form Builder:** The central workspace where fields are added to the form. The fields currently in the form are: Pickup_Address, Contact_Number, Vehicle_Registration_Number, Issue_Description, Odometer_Reading, Preferred_Service_Date, Preferred_Dropoff_Time, Vehicle_Type, Additional_Services_Requested, Multi Select, and a newly added "Authorize_Estimate_And_Repair" field (highlighted in blue).
- Field Properties:** A panel on the right for configuring the selected field. For the "Authorize_Estimate_And_Repair" field, the settings are: Field name: "Authorize_Estimate_And_Repair", Field link name: "Authorize_Estimate_And_Repair", Validation: "Mandatory" (checked), and Data Privacy: "Contains personal data (PII)" (unchecked).

At the bottom of the interface, there is a status bar indicating "Trial expires in 15 days" and an "Upgrade" button. Other navigation links like "My Files", "Chats", and "Contacts" are visible on the left.

Figure 1. Zoho Creator Form Builder showing the Basic Fields and the authorization field configuration.

The screenshot is included as submitted evidence of the Zoho Creator implementation for Experiment 4.

10. Implementation Evidence

Vehicle Service & Repair Request Management

Service_Request_Form

Owner details

Owner_Name *
First Name Last Name

Vehicle_Registration_Number *

Vehicle_Type *
-Select-

contact details

Email_Address

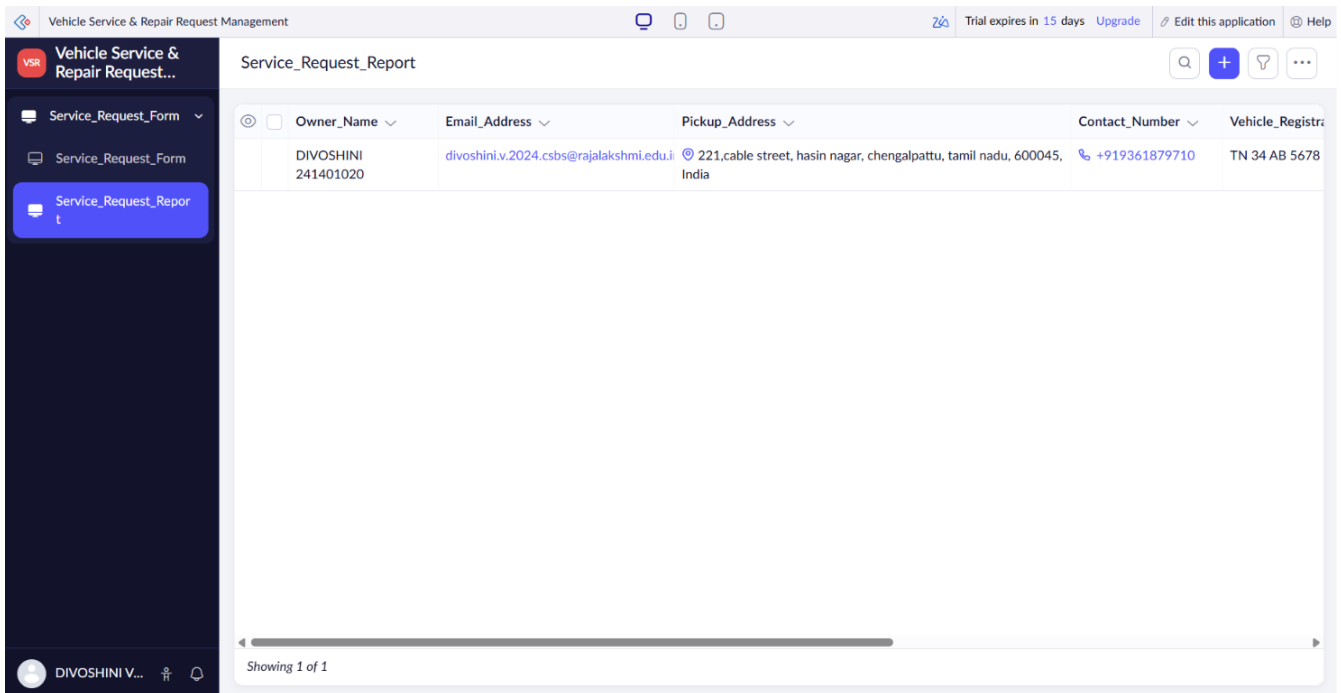
Contact_Number *
+91 81234 56789

Pickup_Address
Address Line 1
Address Line 2
City / District State / Province

Figure 2. Live preview of Service_Request_Form showing the Owner Details and Contact Details sections.

The screenshot is included as submitted evidence of the Zoho Creator implementation for Experiment 4.

10. Implementation Evidence



The screenshot displays a web application interface for 'Vehicle Service & Repair Request Management'. The top navigation bar includes the application title, user profile, and various utility links. The left sidebar contains a menu with options like 'Service_Request_Form' and 'Service_Request_Report', with the latter being the active selection. The main content area, titled 'Service_Request_Report', features a table with one data row. The table columns are Owner_Name, Email_Address, Pickup_Address, Contact_Number, and Vehicle_Registr. The data row contains the following information: Owner_Name (DIVOSHINI 241401020), Email_Address (divoshini.v.2024.csbs@rajalakshmi.edu.i), Pickup_Address (221,cable street, hasin nagar, chengalpattu, tamil nadu, 600045, India), Contact_Number (+919361879710), and Vehicle_Registr (TN 34 AB 5678). Below the table, a pagination bar indicates 'Showing 1 of 1'.

Owner_Name	Email_Address	Pickup_Address	Contact_Number	Vehicle_Registr
DIVOSHINI 241401020	divoshini.v.2024.csbs@rajalakshmi.edu.i	221,cable street, hasin nagar, chengalpattu, tamil nadu, 600045, India	+919361879710	TN 34 AB 5678

Figure 3. Service_Request_Report showing a submitted service request record.

The screenshot is included as submitted evidence of the Zoho Creator implementation for Experiment 4.

11. Expected Learning Outcomes

- Understand the purpose and appropriate use of each Basic Field in Zoho Creator.
- Structure a business form into logical, customer-friendly sections.
- Configure mandatory, format, range and date validations at field level.
- Understand how form values become records in a report and support service scheduling.
- Apply low-code data modelling principles to a realistic service-center workflow.

12. Result

The Vehicle Service & Repair Request Management application was designed using all thirteen Zoho Creator Basic Fields. The form is organized into four logical sections, incorporates the required validations, and feeds submitted records into Service_Request_Report. The design provides a structured pre-service record that can support customer communication, scheduling, technician preparation and service-bay planning.

Conclusion

This experiment demonstrates how data modelling and form design can convert an informal phone-based service process into a consistent low-code business application. By capturing the right information before vehicle arrival and validating it at entry time, the service center can reduce scheduling conflicts and improve operational readiness.